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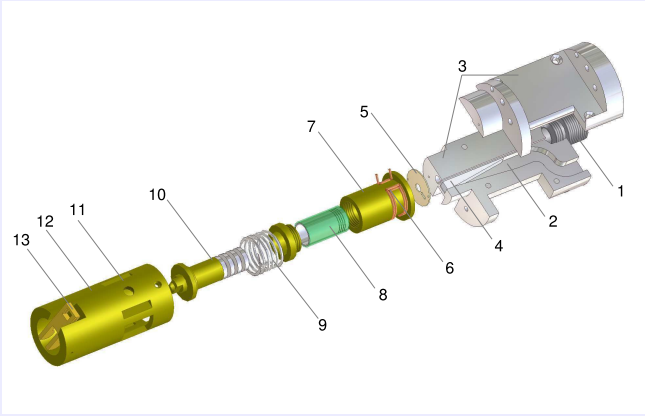
# Universal Probe Head for EPR at 35 GHz

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## Design

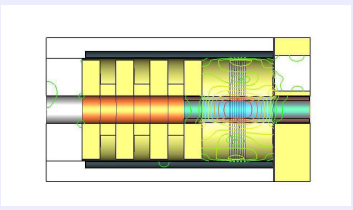
Since Q/Ka band pulse EPR spectrometers became routine spectroscopic tools in physics, chemistry and biology [1-3], the construction of probe heads for daily use on such instruments is a demanding task for the designer. Performance, sensitivity and convenience in use are three major criteria for which an optimal relation has to be reached. We present a probe head which is designed for a 100-W pulse Q-band spectrometer [3] equipped with a CF935 cryostat (Oxford). The probe head is optimized for advanced pulse EPR experiments, pulse ENDOR and cw measurements, with the priority in this order. It is based on a cylindrical cavity (8) coupled on the top to a waveguide via a diaphragm (5) and a Gordon coupling structure (2) with movable teflon wedge (4). The coupling structure has a specially shaped waveguide extruded in a solid block (3), which serves as the base to assemble the holders (7,12), rf coaxes, matching RF circuits and tuning rods. It also provides a guide (1) for the sample holder to allow for a precise positioning of the sample tube (od up to 3.9 mm) into the cavity. The $TE_{011}$ cavity mode is used for the $B_1 \perp B_0$ configuration and the $TE_{112}$ mode could be optionally utilized for $B_1 \parallel B_0$ measurements. However the cavity assembly is optimized to work in the $TE_{011}$ mode. The cavity (8) is made of a quartz tube plated with Au/Ag on its inner surface. On the middle of the resonating volume the metallic layer is circularly scratched to make it transparent to the radio frequency field used for ENDOR experiments. A four-turn saddle coil (6) is wrapped around the cavity housing (7) to produce this field. A movable short (10) allows for a frequency tuning of about 2 GHz for the $TE_{011}$ mode. A lever (13) fixed on a housing (12) and a spring (9) allow the smooth movement of the short (10). Exchangeable modulation coils (not shown) are fixed in slits (11).

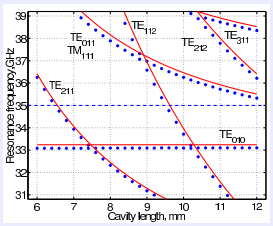
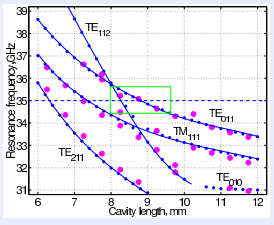
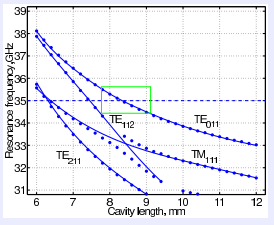
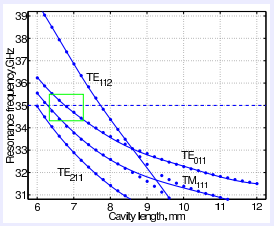
[1] I. Gromov, S. Orlinskii and R. Rakhmatullin, Appl. Magn. Reson, 3,1147 (1992)
[2] C. Davoust, P. Doan and B. Hoffman, J. Magn. Reson. A 119, 38 (1996)
[3] I. Gromov, J. Shane, J. Forrer, R.Rakmatullin, Yu. Rozentzwaig, A. Schweiger, J. Magn. Reson. 149, 196 (2001)




## Simulations

The influence of the coupling structure, the entry holes , and the size and dielectric properties of the sample tube on the cavity eigenmodes and the emf distribution were simulated numerically using the MW Studio package (ver. 4.0, CST). The eigenfrequencies were calculated as a function of the cavity length to simulate the cavity behavior under operation, i.e. when one tries to get the proper frequency at different conditions. Based on these simulations the cavity radius of 5.5 mm has been chosen to avoid a $TE_{011}$ /$TE_{112}$ mode crossing in the working range of our spectrometer (34.5-35.5 GHz). The figures below show the eigenmodes of the cavity. Blue lines and dots represent numerical simulations, red lines are analytical solutions, magenta circles are experimental results, and the green squares are the operational areas in our case. As an example of an emf distribution, the isolines in the figure on the right show the amplitude of the magnetic fields of the $TE_{011}$ mode inside the cavity with an 2.9/2 mm (od/id) empty quartz tube at 35 GHz.

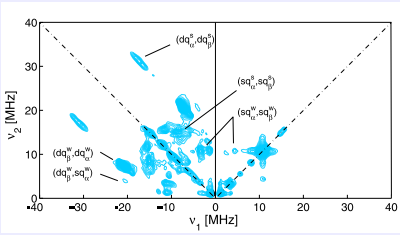


| A. Empty cavity coupled to waveguide, no holes for samples | B. Cavity with 2.9/2 mm (od/id) empty quartz tube, $\varepsilon_t$=3.75 | C. 2.9/2 mm quartz tube, $\varepsilon_t$=3.75, loading $\varepsilon_s$=2 | D. 3.8/2.9 mm empty quartz tube, $\varepsilon_t$=3.75 |
|---|---|---|---|
|  |  |  |  |
| The cavity shows several resonance modes. The numerical solutions (•) agree well with the analytical ones ( ). The $TE_{011}$ and $TM_{111}$ modes are degenerated. The $TE_{212}$, $TE_{311}$ and $TE_{010}$ modes are shown for completeness. These modes do not cross the 35 GHz line under the studied conditions. The $TE_{011}$, $TE_{211}$, $TM_{111}$ and $TE_{112}$ modes have been observed during an exploitaion of the probe. | The $TE_{011}$/$TM_{111}$ degeneracy is lifted due to the sample access holes and the sample tube. All modes are shifted to lower frequencies compared to case (A). The $TE_{112}$ mode is shifted more than the $TE_{011}$ mode. The magenta circles are experimental values. The $TE_{112}$ mode crosses the line of the $TE_{011}$ , but it shows only weak resonances because the coupling structure is optimized for the $TE_{011}$ mode. | When the sample tube is filled with a dielectric material the $TE_{112}$ mode is shifted further down in frequency compared to case (B) and with respect to the $TE_{011}$ resonances. In the case at hand, the $TE_{011}$ mode is always the first observed mode when the cavity length is reduced from 12 to 6 mm using a frequency in the range 33-38 GHz. | The wall of the sample tube in this case is closer to the electric field maxima of the $TE_{011}$ mode than in cases (B,C). As a consequence the $TE_{011}$ mode is shifted more to lower frequencies than the $TE_{112}$ mode. Nevertheless, the $TE_{011}$ mode is well separated from the $TE_{112}$ mode and this situation is not changed when the tube is loaded with dielectric material. |

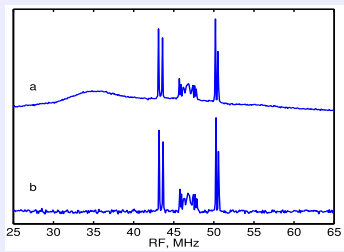
## Performance

| Parameter | Value | Conditions |
|---|---|---|
| Cavity radius, mm | 5.5 | |
| Active length, mm | 6•10 | |
| MW conversion factor, Gauss/$\sqrt{W}$ | 2.5 | Q≈300 |
| RF conversion factor, mT/$\sqrt{W}$ | 0.3 | low Q tuned RF coil |
| Sensitivity, spins/$\sqrt{Hz}$ | $3x10^9$ | pulse mode, RT,Q≈150 |
| Tuning range, GHz | 34-36 | |
| Temperature range, K | 5-300 | |
| Sample access, mm | 3.9 | |
| Q-factor | $10^2$-$10^3$ | |

## Applications




Nitrogen HYSCORE spectrum of the MCRred2 form of methyl-coenzyme M reductase measured at the field position in the middle of the Ni(I) EPR spectrum at 15 K (C. Finazzo et.al., JBIC, 8, 586 (2003)). The single- and double-quantum cross-peaks from the weaker (w) and stronger (s) coupled nitrogens are observed. 16-ns mw pulses have been used to produce π/2 and π pulses in the HYSCORE sequence at 35.3 GHz.




Davies ENDOR spectra of protons weakly coupled to Cu(II) ions doped in a α-glycine single crystal. The spectra were recorded at a temperature of 15 K with an rf pulse length of 25 μs and an unmatched rf coil. The incident rf power was about 450 W. The spectrum (a) was measured in a standard way, i.e. the RF was swept linearly, at a repetition rate of 100 Hz . Because of the weakness of the continuous-flow cooling system the base line is heavily distorted. Applying a *random* RF frequency sweep using recently developed spectrometer software (SpecMan, ETH and WIS, see www.esr.ethz.ch) the baseline distortion is completely eliminated (b).